

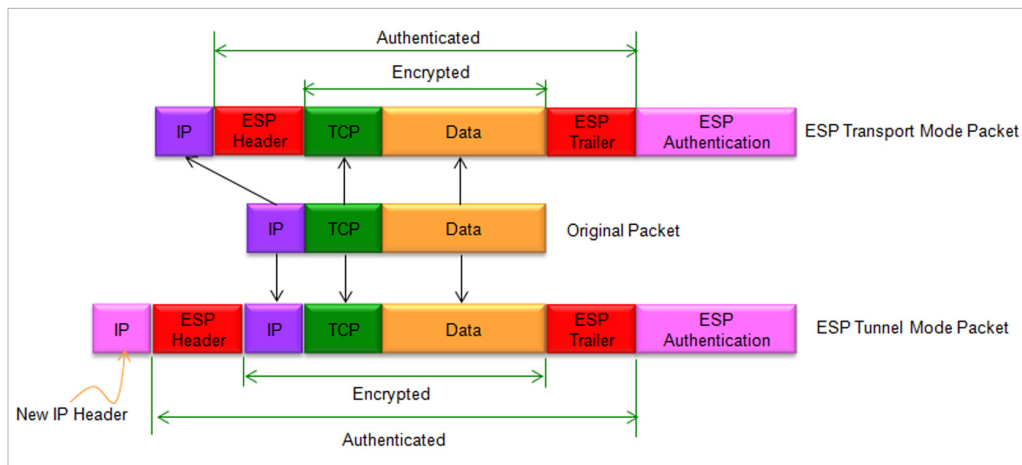


Figure 5: IP security (Source: http://www.sharetechnote.com/image/IP_Security_IPSec_ESP_01.png)

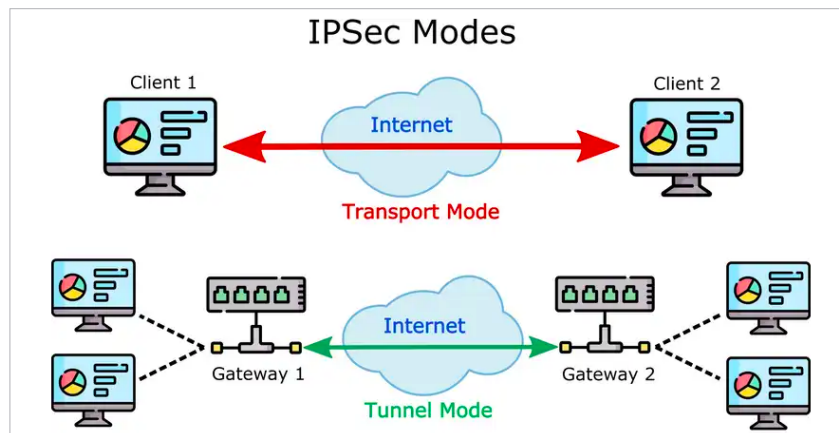


Figure 6: IPSec modes (Source: <https://www.twingate.com/blog/ipsec-tunnel-mode/>)

IKE phase I: In this phase a session is established to negotiate the encryption, authentication and hashing protocols. This collection of protocols is called security association or SA, as mentioned earlier. This phase is used only for the management of traffic.

Here there are two modes. The main mode uses six messages to establish a connection. This mode is more secure but slower. Identification is encrypted. The second mode is the aggressive mode, which uses only three messages to establish connection. It's less secure but the identification is in clear text and faster.

IKE phase II: In this phase, IKE is used to build the connection and

not encrypt user data. AH or ESP is

used to authenticate data and ensure its integrity. Both AH and ESP can also be used at the same time. The data is prepared based on the transport mode. Termination will happen if there is no data transfer for a while.

IPSec passthrough

As we have seen, NAT allows devices to share the same public

IP address. IPSec protocols encrypt the data packets to establish secure connections. This prevents NAT from accessing certain IP header information, like updating the IP source and port numbers, and blocks the packets.

IPSec passthrough allows IPSec tunnels to pass through the router. It safely maintains IP connections over routers that require NAT.

We have learnt about the secure transfer of data until the network layer pretty much in depth. In the next article in this two-part series, we will focus on transport and application layer security. **END** 🐧

References

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The author works in a leading bank as AVP. She is a perennial learner and loves the quote: "The more I learn, the more I realise how much I don't know." - Albert Einstein